

Meta-Learner Selection and Practical Guidance

Sample Size, Imbalance, Regularization Bias and Causal Forests

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Why Meta-Learner Selection Matters

- ▶ Estimating Conditional Average Treatment Effects (CATE) requires balancing:
 - ▶ predictive flexibility with robustness to confounding,
 - ▶ sample efficiency and regularization stability.
- ▶ Different meta-learners behave differently under:
 - ▶ small vs. large samples,
 - ▶ balanced vs. imbalanced treatment assignment,
 - ▶ low- vs. high-dimensional covariates,
 - ▶ weak vs. strong confounding.
- ▶ There is no universally optimal learner:
 - ▶ S-learners emphasize stability and simplicity,
 - ▶ T-learners emphasize flexibility,
 - ▶ X-learners reduce regularization bias,
 - ▶ Causal forests provide local, confounding-robust estimation.
- ▶ Model selection depends critically on data geometry and treatment structure

Scientific Comparison of Meta-Learners

Method	Main Idea	Strengths	Weaknesses	Recommended Setting
S-Learner	Single model $\mu(x, w)$	Efficient use of all data; low variance; simple implementation	Regularization may shrink treatment effects; sensitive to imbalance	Small samples, balanced groups, weak heterogeneity
T-Learner	Separate models for treated/control	Flexible baseline functions; captures structural differences	High variance in small samples; unstable under imbalance	Large balanced samples with heterogeneous response surfaces
X-Learner	Imputed pseudo-effects with propensity weighting	Reduces regularization bias; robust under imbalance	Requires reliable propensity scores; more complex	Moderate/large samples with unequal treatment shares
Causal Forest	Localized regression with adaptive forest weights	Handles confounding, nonlinearities and high dimensions; uncertainty quantification	Computationally intensive; lower interpretability	High-dimensional observational data with complex heterogeneity

Practical Recommendations Summary

Scenario	Recommended Method	Rationale
Small sample ($n < 500$)	S-Learner or X-Learner	Pooling information reduces variance; causal forests may overfit with limited local support.
Moderate sample ($500 < n < 2000$)	X-Learner or Causal Forest	X-learners reduce regularization bias; forests become viable when dimensionality increases.
Large sample ($n > 2000$)	Causal Forest	Adaptive local estimation performs well with sufficient observations per neighborhood.
Strong imbalance ($n_1 \ll n_0$)	X-Learner	Propensity weighting upweights the more reliable pseudo-effect estimate.
High-dimensional covariates ($p > 100$)	Causal Forest	Forests automatically capture nonlinear interactions and heterogeneous treatment structure.
Strong confounding	Causal Forest	Robinson residualization and orthogonalization improve robustness to selection bias.

Decision Rules of Thumb

1. **Start with the data structure:**

- ▶ Small and balanced datasets → S-Learner.
- ▶ Large and balanced datasets → T-Learner.
- ▶ Imbalanced treatment assignment → X-Learner.

2. **Assess dimensionality and confounding:**

- ▶ High-dimensional observational settings strongly favor causal forests.
- ▶ Meta-learners become unstable when nuisance functions are difficult to estimate.

3. **Consider regularization bias:**

- ▶ S- and T-learners may hallucinate heterogeneity when treatment/control regularization differs.
- ▶ X-learners mitigate this by explicitly modeling pseudo-effects.

4. **Prefer orthogonalized methods under confounding:**

- ▶ Robinson-style residualization improves robustness.
- ▶ Causal forests inherit this advantage through localized orthogonal estimation.

5. **No method dominates universally:**

- ▶ Bias–variance trade-offs remain central in CATE estimation.
- ▶ Practical performance depends on overlap, dimensionality, and treatment prevalence.

Key Scientific Takeaways

- ▶ **S-Learners** are variance-efficient but prone to treatment effect shrinkage under regularization.
- ▶ **T-Learners** offer flexibility but require sufficiently large treatment and control samples.
- ▶ **X-Learners** are particularly effective in imbalanced settings because they separate nuisance estimation from treatment effect estimation.
- ▶ **Causal Forests** extend Robinson orthogonalization locally using adaptive forest weights, providing:
 - ▶ robustness to confounding,
 - ▶ nonlinear heterogeneity estimation,
 - ▶ and asymptotically valid inference.
- ▶ Choice should be driven by data geometry rather than algorithmic popularity alone

$$\text{Model selection} \approx f(n, p, n_1/n_0, \text{confounding strength})$$

References I



Athey, S., Tibshirani, J., & Wager, S. (2019).
Generalized Random Forests.
Annals of Statistics, 47(2), 1148–1178.



Chernozhukov, V., Chetverikov, D., Demirer, M., et al. (2018).
Double/Debiased Machine Learning for Treatment and Structural Parameters.
The Econometrics Journal, 21(1), C1–C68.



Künzel, S. R., Sekhon, J. S., Bickel, P. J., & Yu, B. (2019).
Metalearners for Estimating Heterogeneous Treatment Effects using Machine Learning.
Proceedings of the National Academy of Sciences, 116(10), 4156–4165.



Nie, X., & Wager, S. (2021).
Quasi-Oracle Estimation of Heterogeneous Treatment Effects.
Biometrika, 108(2), 299–319.



Robinson, P. M. (1988).
Root- N -Consistent Semiparametric Regression.
Journal of the American Statistical Association, 83(403), 931–938.



Wager, S., & Athey, S. (2018).
Estimation and Inference of Heterogeneous Treatment Effects using Random Forests.
Journal of the American Statistical Association, 113(523), 1228–1242.